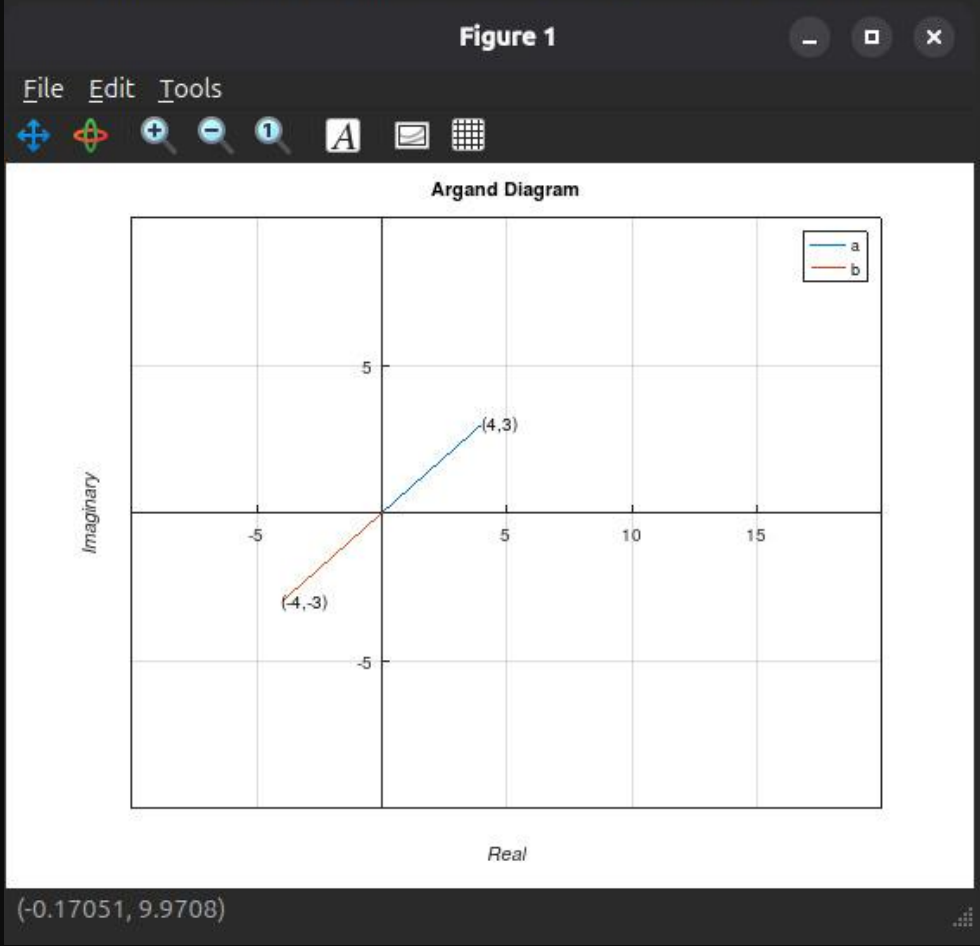
1.3

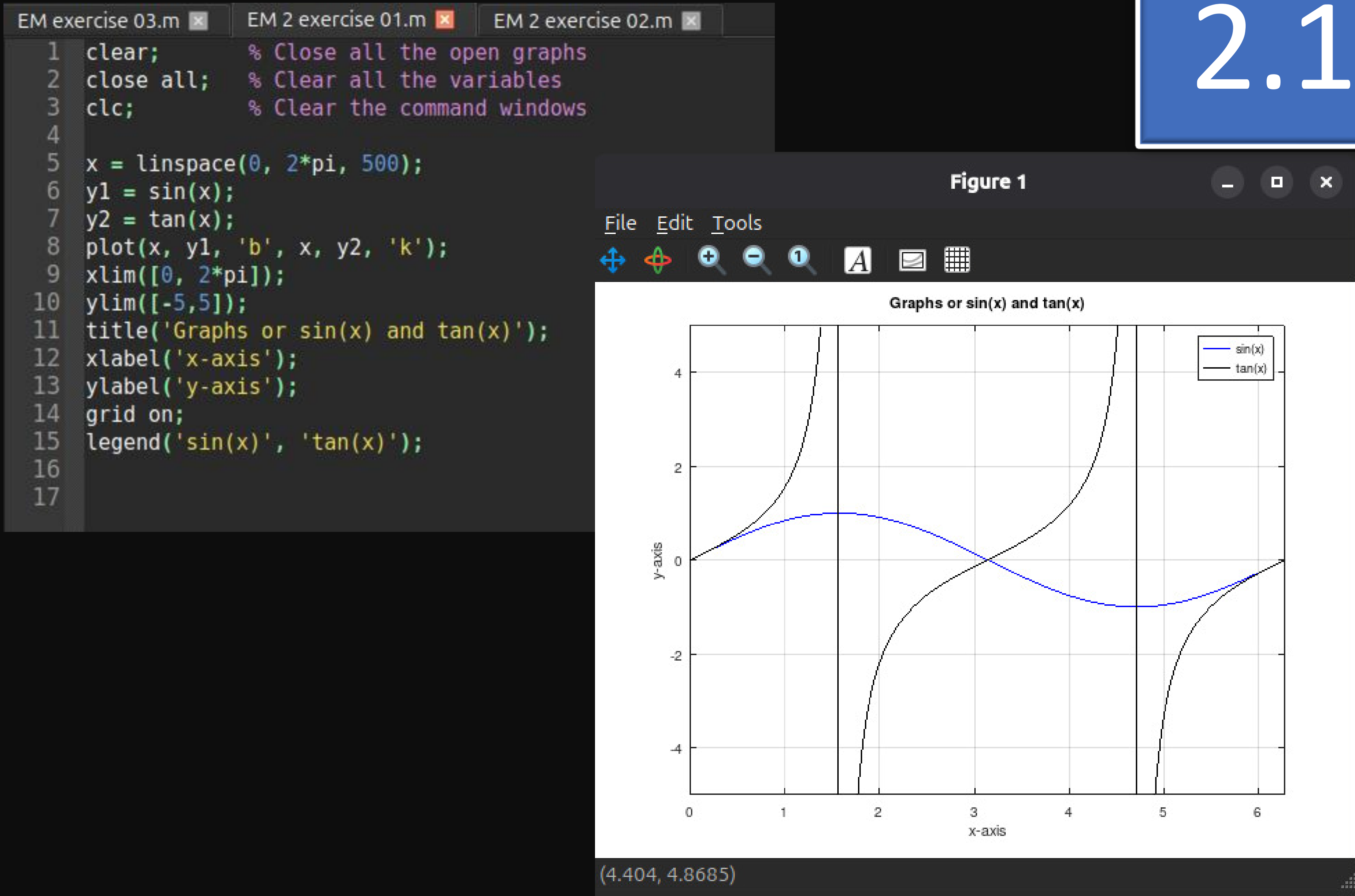
EM exercise 03.m | EM 2 exercise 01.m | EM 2 exercise 02.m

```matlab
clear;        % Close all the open graphs
close all;    % Clear all the variables
clc;          % Clear the command windows

a = 4+3i;
b = -4-3i;

a_x = real(a); a_y = imag(a);
b_x = real(b); b_y = imag(b);

fprintf('a_x = %f\n', a_x);
fprintf('a_y = %f\n', a_y);

printf('Modulus of a =%.2f\n',abs(a)) % Modulus of the complex number a
printf('Argument of a =%.2f degrees\n',(180/pi)*arg(a)) % Argument of the complex number a
printf('Modulus of b =%.2f\n',abs(b)) % Modulus of the complex number b
printf('Argument of b =%.2f degrees\n',(180/pi)*arg(b)) % Argument of the complex number b

figure
plot([0 a_x],[0 a_y],[0 b_x],[0 b_y]) % Calling the plot function.
set(gca, "xaxislocation","origin","yaxislocation","origin") % set the axes acales to go through (0,0).
xlim([-10 20]); ylim([-10 10]); %setting x and y axes limits.
title('Argand Diagram'); % set the title of the graph.
xlabel('\itReal'); ylabel('\itImaginary') % name the axes.
text(a_x,a_y,["(" num2str(real(a)) "," num2str(imag(a)) ")"]) %creat a test label for the coordinate of the complex number
text(b_x,b_y,["(" num2str(real(b)) "," num2str(imag(b)) ")"])
grid('on'); legend('a','b','m=a+b','n=a-b','o=a*b','p=a/b') % turn on the grid lines.
```

```
a_x = 4.000000
a_y = 3.000000
Modulus of a =5.00
Argument of a =36.87 degrees
Modulus of b =5.00
Argument of b =-143.13 degrees
warning: legend: ignoring extra labels.
warning: called from
    legend>parse_opts at line 832 column 9
    legend at line 216 column 3
    /home/senith/Desktop/EM Exercises/EM exercise 03.m at line 27 column 13
```



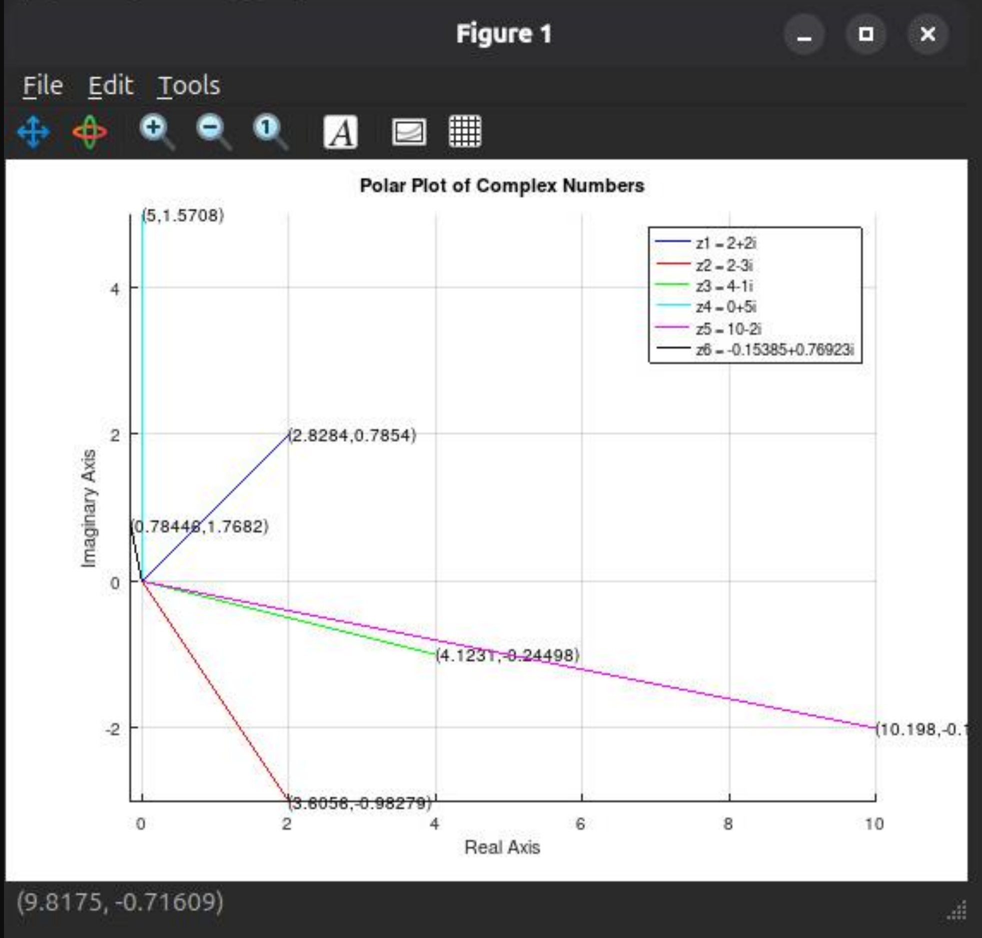
2.1


EM exercise 03.m
EM 2 exercise 01.m
EM 2 exercise 02.m
clear;   % Close all the open graphs
close all;   % Clear all the variables
clc;   % Clear the command windows
x = linspace(0, 2*pi, 500);
y1 = sin(x);
y2 = tan(x);
plot(x, y1, 'b', x, y2, 'k');
xlim([0, 2*pi]);
ylim([-5,5]);
title('Graphs or sin(x) and tan(x)');
xlabel('x-axis');
ylabel('y-axis');
grid on;
legend('sin(x)', 'tan(x)');
Figure 1
File
Edit
Tools
Graphs or sin(x) and tan(x)
sin(x)
tan(x)
y-axis
x-axis
(4.404, 4.8685)

2.2

```
clear;       % Close all the open graphs
close all;   % Clear all the variables
clc;         % Clear the command windows

a = 2 + 2i;
b = 2 - 3i;

m = a+b;
n = a-b;
o = a*b;
p = a/b;

%Modulus and argument of each value
abs_a = abs(a); arg_a = arg(a);
abs_b = abs(b); arg_b = arg(b);
abs_m = abs(m); arg_m = arg(m);
abs_n = abs(n); arg_n = arg(n);
abs_o = abs(o); arg_o = arg(o);
abs_p = abs(p); arg_p = arg(p);

printf('m = %.0f%+.0fi\n', real(m), imag(m));
printf('n = %.0f%+.0fi\n', real(n), imag(n));
printf('o = %.0f%+.0fi\n', real(o), imag(o));
printf('p = %.4f%+.4fi\n', real(p), imag(p));

printf('a_mod : %.3f a_angle : %.3f\n', abs(a), arg(a));
printf('b_mod : %.3f b_angle : %.3f\n', abs(b), arg(b));
printf('m_mod : %.3f m_angle : %.3f\n', abs(m), arg(m));
printf('n_mod : %.3f n_angle : %.3f\n', abs(n), arg(n));
printf('o_mod : %.3f o_angle : %.3f\n', abs(o), arg(o));
printf('p_mod : %.3f p_angle : %.3f\n', abs(p), arg(p));

z = [a, b, m, n, o, p];
colors = ['b', 'r', 'g', 'c', 'm', 'k'];
%Blue, Red, Green, Cyan, Magenta, Black
```

```
m = 4-1i
n = 0+5i
o = 10-2i
p = -0.1538+0.7692i
a_mod : 2.828 a_angle : 0.785
b_mod : 3.606 b_angle : -0.983
m_mod : 4.123 m_angle : -0.245
n_mod : 5.000 n_angle : 1.571
o_mod : 10.198 o_angle : -0.197
p_mod : 0.784 p_angle : 1.768
```



```
figure;
hold on;     %keep all 6 lines on the same axes
axis equal;  %show gridlines for easier reading
grid on;     %equal scaling on both axes so angles look correct

for k = 1:length(z)
  x = real(z(k));    %horizontal component (real part)
  y = imag(z(k));    %vertical component (imaginary part)
  plot([0 x], [0 y], colors(k)); %draw a line from the origigin (0,0) to the point (x,y)
% label the tip of the vector with its modulus and angle (in radians)
  text(x, y, ["(" num2str(abs(z(k))) "," num2str(arg(z(k))) ")"]);
end
title('Polar Plot of Complex Numbers');
xlabel('Real Axis');
ylabel('Imaginary Axis');
legend(
         ['z1 = ' num2str(a)],
         ['z2 = ' num2str(b)],
         ['z3 = ' num2str(m)],
         ['z4 = ' num2str(n)],
         ['z5 = ' num2str(o)],
         ['z6 = ' num2str(p)]
      );
hold off;

```